

Moving from the thermal infrared (LWIR) down to the shortwave infrared solar reflection spectrum (SWIR, roughly 1.0 to 3.0 μm), the fundamental SARTA predictor variables (T , q , θ) remain the same, but the **underlying radiative physics changes completely**.

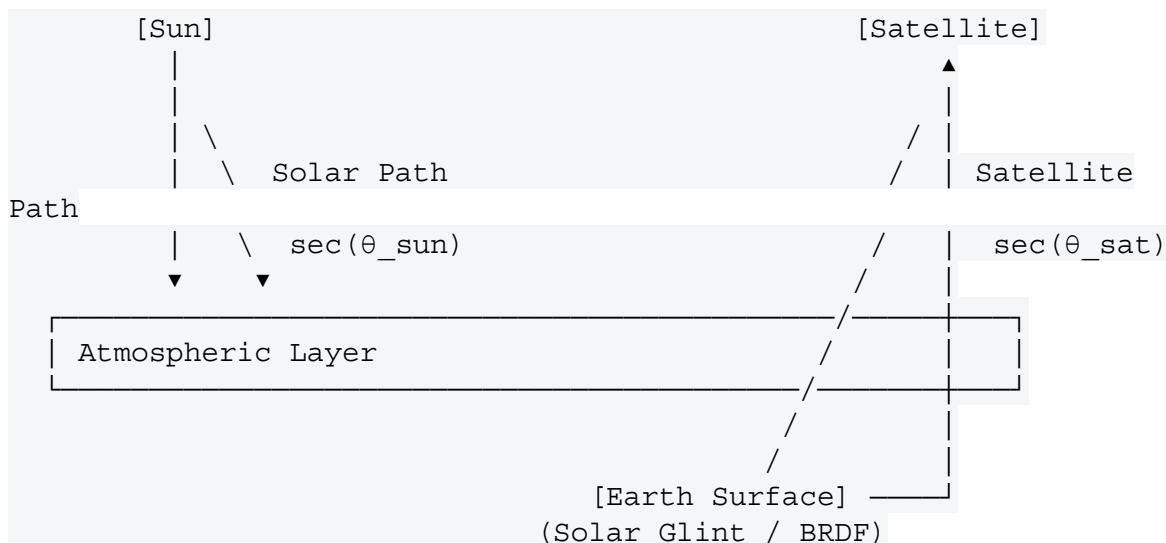
To account for these physical shifts, SARTA adapts its predictor loops and integration steps through three key changes:

1. Dual-Path Geometry Predictors (The Solar Slant Path)

In the thermal infrared spectrum, SARTA only tracks a single path: the upwelling path from the Earth's surface straight to the satellite sensor.

In the shortwave solar spectrum, daylight enters from the Sun, travels down through the atmosphere, reflects off the surface, and travels back up to the satellite. SARTA accounts for this dual-path geometry by updating its path length predictor variables:

- **Thermal Path:** $\sec(\theta_{\text{sat}})$, where θ_{sat} is the satellite viewing zenith angle.
- **Solar Path Modifier:** SARTA adds $\sec(\theta_{\text{sun}})$ to its predictor equations, where θ_{sun} is the Solar Zenith Angle.
- **Combined Predictors:** Predictor terms that previously used $\sec(\theta)$ are expanded to incorporate the total two-way path factor, $(\sec(\theta_{\text{sat}}) + \sec(\theta_{\text{sun}}))$, to capture how much more gas absorption occurs along the solar beam's journey.



2. Shifts in Molecule Sensitivity and Line Saturated Terms

The way gases absorb light changes significantly between the longwave and shortwave

spectrums, which alters how SARTA weighs its polynomial terms:

- **Water Vapor Continuous Absorption:** In the longwave spectrum, water vapor continuum absorption (molecules colliding with each other) is a dominant factor, requiring heavy use of the squared terms $(W_r \cdot \sec(\theta))^2$. In the shortwave spectrum, line absorption dominates over continuum absorption, shifting the mathematical weight toward the linear and square-root terms ($\sqrt{W_r}$).
- **Trace Gas Dominance:** Shortwave channels are highly sensitive to gases like **Carbon Monoxide (CO)** and **Methane (CH₄)**. SARTA scales up its independent trace gas predictor loops in this spectrum to track localized fluctuations in these specific molecules.

3. Modifying the Boundary Equation

When integrating the final top-of-atmosphere radiance, SARTA switches its lower boundary calculation from a simple thermal blackbody emission to a **solar specular reflection** model:

- **Deactivating the Planck Function:** SARTA deactivates the standard thermal Planck function $B(\lambda, T_{\text{skin}})$ for these channels because the Earth's surface does not emit meaningful thermal energy at wavelengths shorter than 3 μm .
- **Activating Surface Albedo (BRDF):** SARTA replaces thermal emissivity (ϵ) with a Bidirectional Reflectance Distribution Function (BRDF). The input solar beam is multiplied by this surface reflectivity and then attenuated by the dual-path gas transmittances calculated from the updated predictor equations.

Now that you have explored both the longwave and shortwave frameworks, we have covered the complete forward-model architecture of SARTA. Would you like to wrap up by seeing how this entire system is deployed in **real-time weather forecasting systems** like NOAA's NCEP or ECMWF?